Supplementary material for "Is the closing-auction imbalance already in the price?"

This supplement adds the figures and tables behind every check in the main report. Every number is generated by the same code run as the main report. Section numbers here are prefixed with S.

S1. Data processing

The hidden index

The target is a stock's 60 s WAP return minus an index's 60 s return, and the index is not given. At every timestamp the implied index return (stock return minus target) is the same for all stocks to within 0.005 bps, so it is one series. Regressing it on the 200 stock returns recovers positive weights that sum to 1.0000 and leave unexplained variance 2.1×10^{-8}. The weights come in a few discrete tiers (Figure S1), the largest at 0.040.

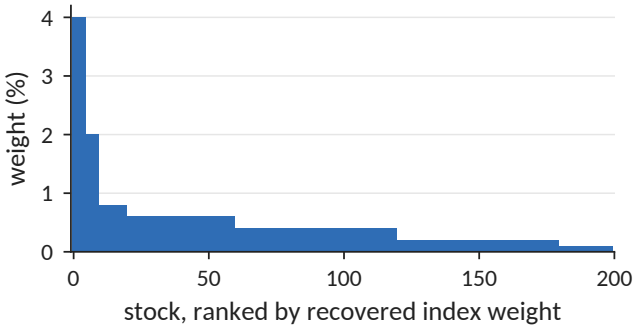


Figure S1: The hidden index is a tiered weighted average of the 200 stocks. Bars are the recovered weights, largest first.

Streaming summary

A Count-Min sketch with 7 rows of 2,000 counters ($\varepsilon = 0.001$, $\delta = 0.01$, universal hashing modulo the prime $2^{31} - 1$) totalled the imbalance shares of 95,232 stock-days in one pass. It uses 14,000 counters, 15% of exact counting. The guarantee says each estimate exceeds the truth by at most εN with probability at least $1 - \delta$; here 100% of estimates do, the worst at 66% of the bound (Figure S3).

S2. Distributions

The t fit raises the log-likelihood by 0.125 per row and cuts the 20-bin χ^2 goodness-of-fit statistic from 98,064 to 568. With fewer than four degrees of freedom the fourth moment is infinite, which is why the sample kurtosis (22.6) is unstable and why our tests rely on day averages.

A stock changes imbalance side 2.69 times per auction on average, with variance 6.59. A Poisson has variance equal to its mean, so the dispersion of 2.45 rejects it ($\chi^2 = 189{,}527$ on 10 d.f.).

S3. P2: learned models

Setting	Value, fixed before training
Split	train days 0 to 240 (2,600,950 rows), early stopping on 241 to 300, test on 301 to 480
Features	signed imbalance, |imbalance|/matched, side, log matched size, spread, near and far gaps with missing flags, reference-to-WAP gap, depth imbalance, time, phase, WAP returns over 10, 30, 60 s, imbalance changes over 10, 60 s; no stock identity
MLP	128 and 64 ReLU units, dropout 0.1, Adam at 10^{-3}, batch 4,096, MSE on the winsorised target, early stopping on validation IC
LightGBM	63 leaves, learning rate 0.05, early stopping after 100 rounds
GRU	one layer of 64 units over the auction so far, snapshot features only
Seeds	5 per model; predictions averaged

Comparison (held-out days)	Estimate (95% CI)
MLP minus imbalance, WAP target	$+0.140$ (0.134 to 0.146)
LightGBM minus imbalance	$+0.148$ (0.142 to 0.153)
MLP minus LightGBM	-0.008 (-0.009 to -0.006)
GRU minus snapshot-only MLP	$+0.022$ (0.020 to 0.025)
MLP minus imbalance, mid target	-0.010 (-0.016 to -0.003)
Diebold-Mariano, MLP vs zero	$t = 17.1$
Diebold-Mariano, MLP vs linear	$t = 16.9$
Seed SD of mean IC (MLP, LGBM, GRU)	0.0010, 0.0004, 0.0022

The GRU's gain over the snapshot-only MLP says the path of the auction carries information beyond its current state, on the WAP target. Seed-to-seed variation is two orders of magnitude below every gain (Figure S6), so none of these differences is training noise.

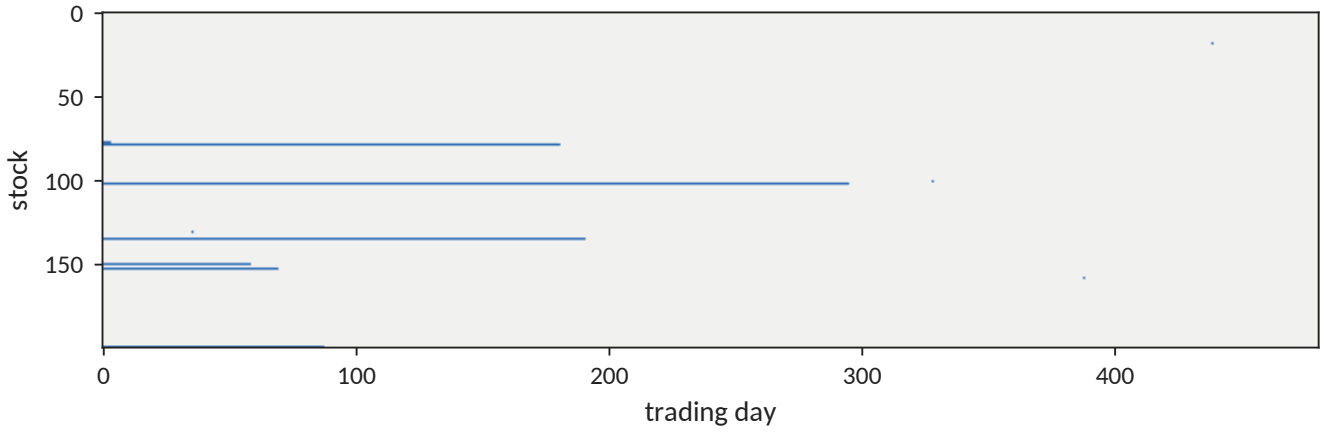


Figure S2: Panel coverage: each row is a stock, each column a trading day, and blue marks a stock-day with no data. 968 of 96,200 stock-days are absent, almost all from 15 stocks that join the panel late; one stock is missing for 295 days. Every comparison in the report is made within a day.

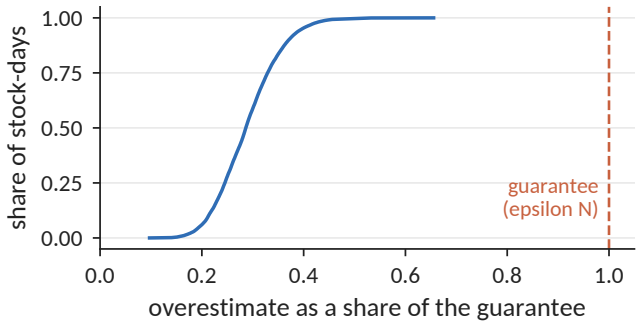


Figure S3: Every Count-Min estimate stays inside its guarantee. The curve is the cumulative share of stock-days whose overestimate is below a given fraction of εN.

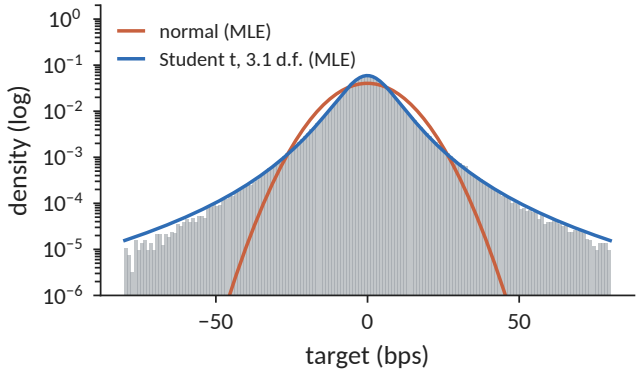


Figure S4: A Student t with 3.1 degrees of freedom fits the tails that a normal misses. Both curves are maximum-likelihood fits to the minute snapshots; grey is the histogram.

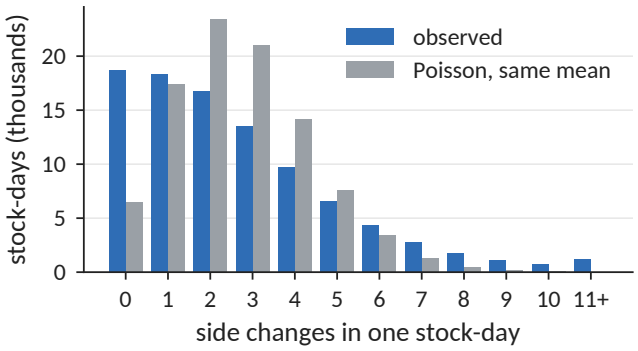


Figure S5: Auctions either hold one side all along or churn: far more stock-days have no side change, and more have many, than a Poisson count with the same mean allows.

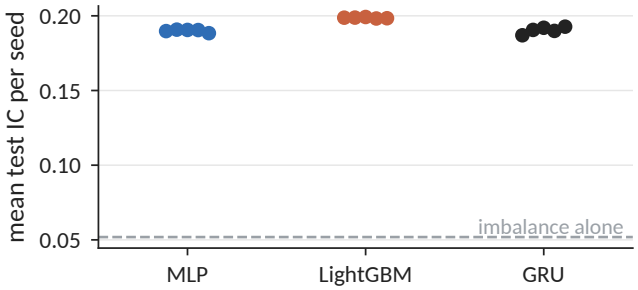


Figure S6: Five seeds per model agree almost exactly, far above the imbalance alone on the WAP target.

S4. P3: placebo, subgroups and the classifier test

Subgroup	IC pre	IC post	△IC (95% CI)
first half of days	0.024	0.087	+0.063 (0.054 to 0.072)
second half of days	0.041	0.073	+0.031 (0.022 to 0.041)
size: low	0.050	0.080	+0.029 (0.021 to 0.037)
size: mid	0.033	0.075	+0.041 (0.032 to 0.051)
size: high	0.016	0.098	+0.081 (0.071 to 0.091)
spread: low	0.016	0.094	+0.078 (0.069 to 0.088)
spread: mid	0.035	0.073	+0.038 (0.028 to 0.047)
spread: high	0.046	0.082	+0.036 (0.028 to 0.043)
day volatility: low	0.038	0.077	+0.039 (0.029 to 0.050)
day volatility: mid	0.032	0.080	+0.048 (0.036 to 0.059)
day volatility: high	0.028	0.082	+0.055 (0.041 to 0.068)

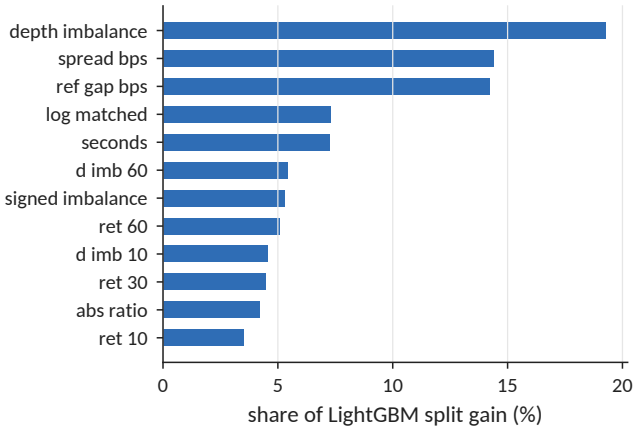


Figure S7: The models lean on book depth and the spread, the inputs that move the WAP mechanically, more than on the imbalance (5% of split gain).

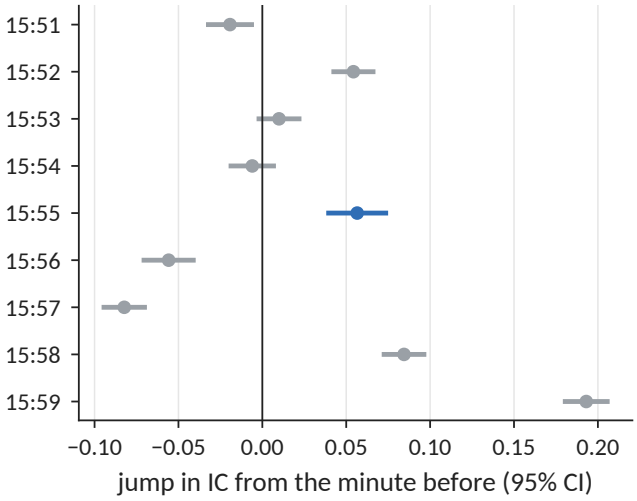


Figure S8: The jump at 15:55 is real but not unique: the last minute jumps more. Each line is the change in IC from the previous minute, with a 95% interval over days.

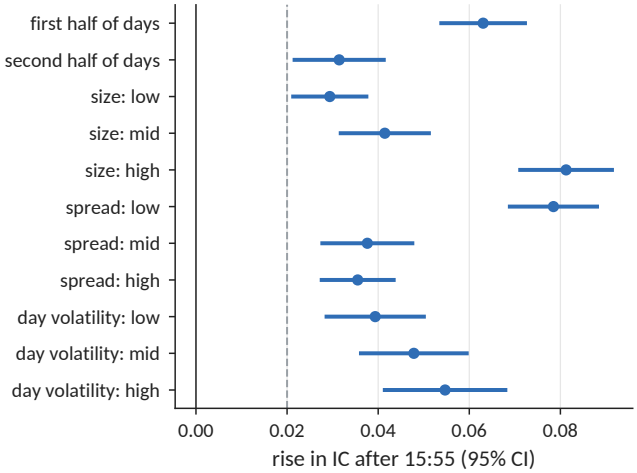


Figure S9: The post-15:55 rise holds in every subgroup fixed in advance. Dashed line: the 0.02 threshold.

Classifier two-sample test

If the auction state did not change at 15:55, no classifier could tell pre from post snapshots better than chance. LightGBM, trained without the features that are missing by rule, is right on 82.3% of 359,960 held-out snapshots ($z = 388$; day-level 95% CI 82.0 to 82.7%). This is the lecture 6 one-proportion z-test applied to a learned classifier's accuracy.

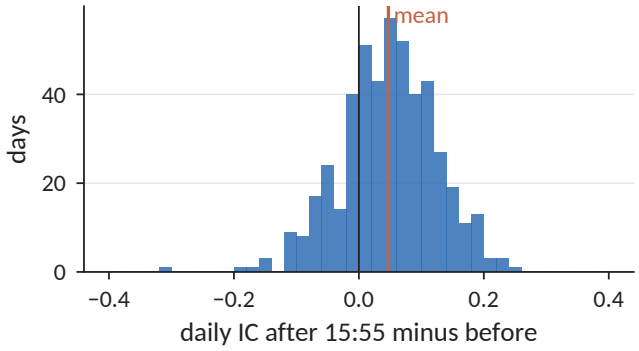


Figure S10: Day by day, the imbalance predicts better after 15:55 on 363 of 481 days.

S5. P4: edge, cost and the learned rule

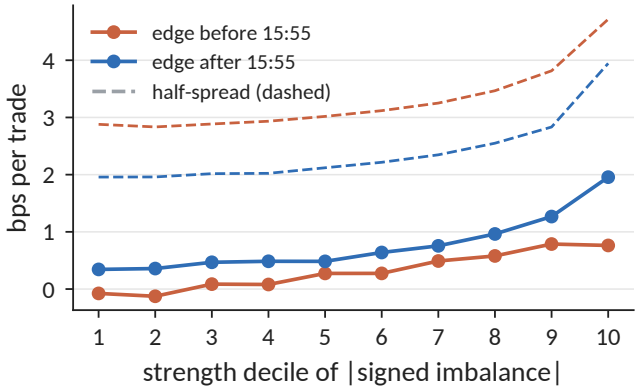


Figure S11: Stronger imbalances earn more, but their stocks quote wider spreads, so the edge never catches the cost. Solid: gross edge by strength decile; dashed: the same stocks' half-spread.

Rule (held-out days)	Net bps per trade (95% CI)
MLP signal above half-spread, WAP target	$+0.68$ (0.58 to 0.77), hit rate 53.5%
Same rule, mid-price target	-0.40 (-0.51 to -0.28), hit rate 46.8%
Imbalance, 15:59 snapshot only	$+0.48$ (0.39 to 0.58)
first half of days	$+0.74$
second half of days	$+0.23$ (0.10 to 0.36)

S6. Power

Test	Detectable at 80%	Threshold	Days needed at threshold
Test 1	0.036 bps	1.85 bps	1
Test 4	0.010 IC	0.02 IC	115
Test 5	0.027 log ratio	0.105	32
Test 6	0.002 hit rate	0.02	5

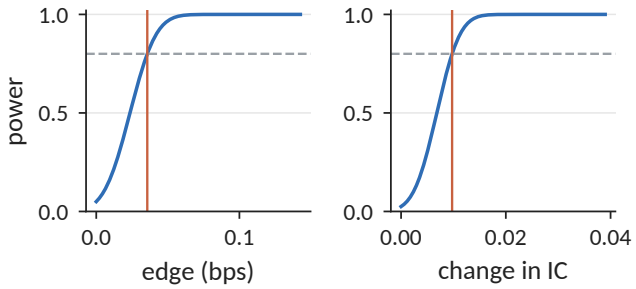


Figure S12: With 481 days, power reaches 80% (dashed) at effects far below our thresholds. Left: Test 1; right: Test 4. The vertical line marks the detectable effect.

S7. Pre-registration record

Every threshold was written down, with a date, before the analysis it governs was run. Changes made before any test ran, and results that missed their pre-set criterion, are listed here.

When	What was fixed or changed
Before tests	sampling at whole minutes; the SDM as the median half-spread; Tests 1, 2, 4, 5, 6 with their thresholds
Before tests	a far-price fence changed from Tukey (flags 26%) to Chebyshev (flags 207 rows); a cleaning step dropped as a profiling artefact
Before checks	subgroups, placebo, power, cost, permutation, fits and sketch criteria
Before training	Test 3 design, models, split and checks
Before 5 seeds	the mid-price check, added after a one-seed run whose numbers are not reported
Miss	Test 4: we predicted IC would fall after 15:55; it rose
Miss	Placebo: 15:55 is not the largest jump
Rule fired	Test 3: most of the learned gain is WAP mechanics

S8. Reproduction

One command, `python run_all.py`, rebuilds every number and figure in both documents from the raw `train.csv` in about seven minutes on a laptop with a consumer GPU. Every number in the text is a macro written by the code; none is typed by hand.